

A Project Report On Nepal Census 2011

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Abstract

This project presents a comprehensive data-driven analysis of Nepal's 2011 national census, aiming to extract meaningful insights from demographic, social, and economic data. Utilizing detailed datasets published by the Central Bureau of Statistics (CBS), the study examines various dimensions of the population, including age distribution, gender ratio, educational status distribution, employment status, household characteristics, and geographic patterns. A rigorous data science pipeline was implemented, encompassing data cleaning, transformation, exploratory analysis, and visualization using Python and its scientific ecosystem. By uncovering spatial and temporal patterns, the analysis highlights regional disparities in development indicators, urban-rural divides, and key challenges in education, employment, and infrastructure. The study also identifies clusters of underdevelopment and population pressure, providing evidence-based insights for targeted policy interventions. The findings offer valuable support for planners, policymakers, and development practitioners, demonstrating how data analytics can inform sustainable development and equitable resource distribution across Nepal. This work showcases the critical role of census data in guiding national strategies and fostering inclusive growth.

1. Introduction

Data is crucial for understanding and addressing complex social, economic, and demographic challenges. In developing countries like Nepal, national census data serves as a vital source of information for informed decision-making and policy formulation. This report presents a comprehensive data science approach to analyzing the 2011 Nepal Census, applying exploratory data analysis (EDA) and a range of analytical techniques to uncover key insights into the country's population dynamics, socio-economic indicators, and regional disparities.

2. Project Overview

Objective

To extract meaningful and actionable insights from the 2011 Nepal Census data using comprehensive exploratory data analysis (EDA) and advanced data science techniques.

Scope

- In-depth analysis of Nepal's 2011 census dataset
- Identification of demographic, socio-economic, and regional patterns
- Exploration of variables such as population distribution, literacy, employment, housing, and migration
- Generation of visual and statistical summaries to support data-driven policy formulation and academic research

Methodology

- **Programming Language:** Python
- **Libraries:** pandas, numpy, matplotlib, seaborn, plotly, pyreadstat
- **Pipeline:** Data Cleaning → Exploratory Data Analysis → Visualization → Modeling

Questions You Can Ask With This Data

Descriptive Questions

1. Population and Household Distribution

- How many households are there across the country and in each district?
- How many VDCs/Municipalities exist in each district, and how many of them are Urban or Rural?
- What is the total number of Urban and Rural areas in the country?

2. Absentee Population Analysis

- How many households reported having absentee members across different districts?
- What is the distribution of absentees by sex, age group, and reasons for absenteeism?

3. Housing Characteristics and Ownership

- What are the predominant types of house construction (foundation, wall, and roof materials) across different ecological belts?
- What are the types of house ownership across the country, and how are houses distributed by roofing and wall material, water source, electricity source, and types of cooking fuel?
- What is the status of women's house ownership at national and district levels?
- Which districts have the highest number of female-headed land ownerships?

4. Mortality and Disability Statistics

- What is the total number of deaths in the country and in each district?
- What is the distribution of deaths by age group and gender?
- What is the district-wise mortality rate?
- What are the types and numbers of disabilities reported in the country?

5. Education, Religion, and Employment

- What is the distribution of individuals by level of education and religion?
- How is occupation and employment status distributed among the population?

Diagnostic Questions

1. Why are absentees counted across different age groups?
2. What are the Absentees by destination rate?
3. What are the Reasons of absentees by Destination?
4. What are the Distribution of House Ownership Type by District.
5. What might explain the low rate of female land ownership in specific regions and which districts have higher female ownership rate?
6. Can foundation type of household be affected by type of ownership?

Predictive Questions

1. Can we predict the Reason of Absence based on Age, Sex, Education, and District?
2. Can we predict absenteeism likelihood based on household characteristics like education, ownership, or housing type?
3. Can we make association rules for house characteristics and ownership?
4. Can we predict Which types of owner according to the house feature ?

Prescriptive Questions

1. What policy interventions could reduce absenteeism or better support absentee households?
2. What kind of housing improvement programs should be prioritized in rural versus urban regions?
3. Which districts should be targeted for increasing female land ownership through government schemes?

Comparative Questions

1. How does absenteeism differ between ecological zones (Terai vs. Hill vs. Mountain)?
2. How does the household death rate correlate with urban vs. rural areas?
3. Is there a notable difference in house ownership patterns across development regions?
4. How do the education levels of absentees compare by gender and region?

3. Data Introduction

The 2011 Nepal Census, conducted by the **Central Bureau of Statistics (CBS) Nepal**, is a nationwide demographic and socio-economic survey covering over 5.4 million households. It provides critical data on population distribution, housing, migration, education, and land ownership across all 75 districts, serving as a foundational source for policy and development planning in Nepal.

The dataset is distributed in six different parts, given the names (Absentee, Household, BatchId, Death, Individual01, and Individual02)

1. Absentee: This dataset captures detailed records of household members who were absent from their homes during the 2011 census, primarily due to foreign employment, education, or other reasons. It enables granular analysis of labor migration trends, demographic characteristics of absentees, and their destinations, which are critical for understanding socio-economic mobility and regional labor patterns.

Shape: ~2,354,166 rows $\times$ 13 columns

Key Features: District, V.D.C./Municipality, Ward No., Enumeration Area, Household & House No., Absentee ID, Sex, Age, Education, Duration of Absence (in years), Reason for Absence, Destination Country.

2. Batch (Regional Metadata) Dataset: This metadata provides the geographic and administrative classification framework for all surveyed regions in the 2011 Nepal Census. It is essential for stratifying other census datasets by region, enabling comparative analysis across Nepal's development regions, ecological belts, and urban-rural divisions.

Shape: ~3,915 rows $\times$ 9 columns

Key Features: District Code & Name, V.D.C./Municipality Code & Name, Combined Administrative Code, Development Region, Ecological Belt, Eco-Development Region, Urban/Rural Classification

3. Household Dataset: This dataset provides an extensive view of household-level demographics, housing infrastructure, and land ownership in Nepal. It serves as a foundational source for analyzing socio-economic conditions, gender disparities in land

ownership, housing quality, and the direct impact of migration and mortality on family units.

Shape: ~5,830,000 rows $\times$ ~30+ columns (varies slightly by format)

Key Features: District, V.D.C./Municipality, Ward, Household ID, Housing Characteristics (ownership, foundation, wall, roof types), Land Ownership by Women (in Bigha, Kattha, Dhur, Ropani, Aana, Paisa), Household Mortality (deaths in past 12 months), Absentee Presence & Count

4. Data Exploration Techniques

4.1. Descriptive Statistics

Descriptive statistics were fundamental in summarizing the key characteristics of the Nepal Census 2011 dataset. Measures such as mean, median, mode, and standard deviation were calculated for variables like household size, age, literacy rate, and income-related proxies. These statistics provided a foundational understanding of central tendencies and variability across the population, helping to highlight socio-economic disparities across different regions, castes, and ethnicities. Such descriptive insights were essential for setting the context before deeper analyses were conducted.

4.2. Distribution Analysis

To further understand how data values were spread across the population, distribution analysis was performed using visual tools like histograms and boxplots. These helped reveal skewness, concentration, and outliers in variables such as income, age, and educational attainment. Urban and rural comparisons showed distinctly different distribution patterns, with urban areas tending toward lower variability in income and better access to education. Boxplots also helped highlight inter-group disparities in health, literacy, and household assets, uncovering regional and social inequalities.

4.3. Grouping & Aggregation

Grouping and aggregation techniques were applied to draw meaningful summaries from the diverse census data. Data was grouped by administrative levels such as district, municipality, and ward, as well as social classifications like caste, ethnicity,

and gender. Aggregated indicators such as total population, literacy rate, average household size, and gender ratios were calculated to compare trends across regions and demographic groups. These aggregated statistics allowed for an efficient comparison of social and economic development across different parts of the country.

4.4. Correlation Analysis

To identify relationships between variables, both Pearson and Spearman correlation analyses were conducted. These correlations uncovered significant associations, such as between literacy rate and employment type, or female education and household income proxy indicators. Strong correlations were also found between access to services like electricity and education levels. These insights helped in identifying key determinants of socio-economic status and guided the formulation of hypotheses for further investigation.

4.5. Outlier Detection

Outlier detection was carried out using statistical methods such as the Z-score and interquartile range (IQR). These methods helped identify unusually high or low values in key variables like population count, household income, and literacy rates at local administrative levels. Some regions displayed extreme values, which were further examined to determine if they were due to data entry errors or reflected genuine anomalies. Detecting and understanding these outliers was essential for improving data quality and drawing accurate conclusions.

5. Data Preprocessing Steps

5.1. Data Cleaning

Data cleaning involved handling missing values, removing duplicates, and resolving inconsistencies across variables. Missing values were addressed using appropriate techniques such as median imputation or logical inference, especially in fields like household amenities or individual characteristics. Duplicates were identified and removed, ensuring data integrity. Inconsistent naming and coding—especially in categorical variables such as religion, caste, and location—were standardized to enable accurate grouping and analysis.

5.2. Data Transformation

Transformation techniques were implemented to prepare the data for advanced analysis and machine learning tasks. Numerical features were normalized and scaled to ensure uniformity, particularly for clustering and classification tasks. Categorical variables such as education level, religion, and housing type were encoded using techniques like label encoding and one-hot encoding. These transformations ensured that the dataset could be efficiently processed by algorithms and allowed meaningful comparisons across groups.

5.3. Feature Engineering

New features were derived from existing variables to enrich the dataset and support deeper insights. Notable engineered features included the dependency ratio, which measured the proportion of dependents to working-age individuals, and female land ownership ratio, which provided a gender-based economic indicator. Migration history was used to generate time-lagged features for analyzing population movement trends. These newly created features enabled more granular socio-economic analysis and revealed hidden patterns not directly visible in raw data.

6. Exploratory Data Analysis (EDA)

6.1. Global and Thematic EDA

Exploratory data analysis at a national level uncovered major patterns in Nepal's demographic and socio-economic structure. Urban centers like Kathmandu showed higher literacy and income levels, while remote districts exhibited lower access to services. Ethnic and caste disparities were evident in education, employment, and land ownership. Internal migration trends revealed significant movement from rural to urban areas and from hills to the Terai, driven by education, employment, and access to services.

6.2. Nepal-Specific EDA

The EDA also included Nepal-specific insights, such as district-wise analysis of literacy and gender ratios, variations in female land ownership, and caste-based disparities in educational attainment. Comparative analyses between rural and urban regions highlighted the urban-rural divide in access to electricity, sanitation, and education. Migration data provided insights into labor migration trends and

their social impact, such as the rise in female-headed households and changing household compositions.

6.3. Visualizations

Visualizations played a key role in interpreting and communicating the results of the analysis. Line plots were used to display migration and age distribution trends, while bar charts illustrated literacy rates, employment types, and housing conditions by district. Heatmaps revealed correlations between variables like education, income proxies, and infrastructure access. Geospatial maps developed using GeoPandas and Folium provided a visual representation of development indicators across districts, helping to pinpoint regions in need of targeted interventions.

7. Data Visualization Techniques

7.1. Tools

- **Matplotlib** and **Seaborn** for static plots
- **Plotly** for interactive dashboards (optional)

7.2. Key Visualizations

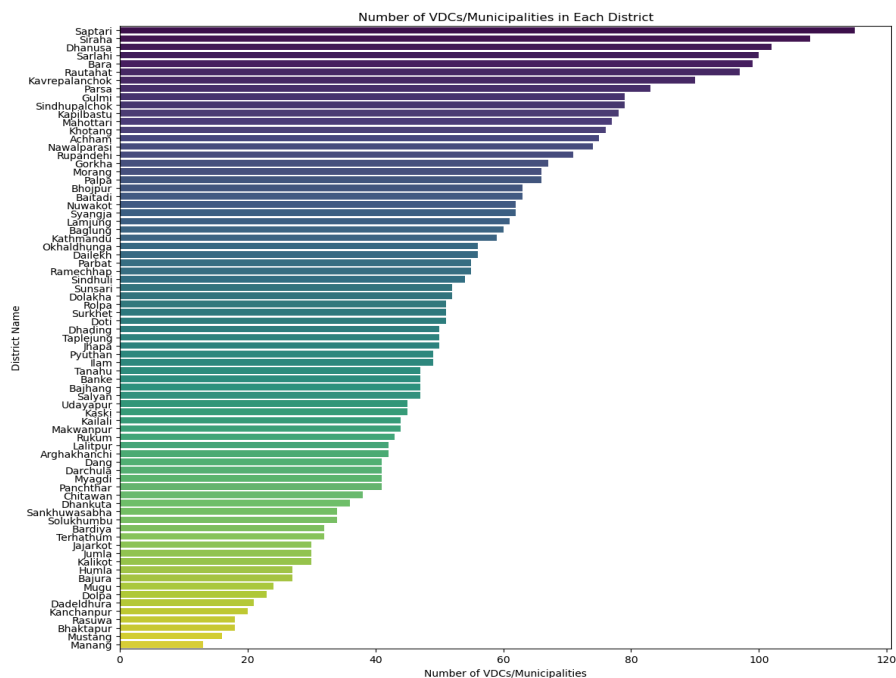
- **Time Series Plots:** Production and yield over the years
- **Bar Charts:** Top commodities by value/volume
- **Heatmaps:** Correlation between variables
- **Geospatial Maps:** Regional production patterns within Nepal

8. Key Insights & Findings

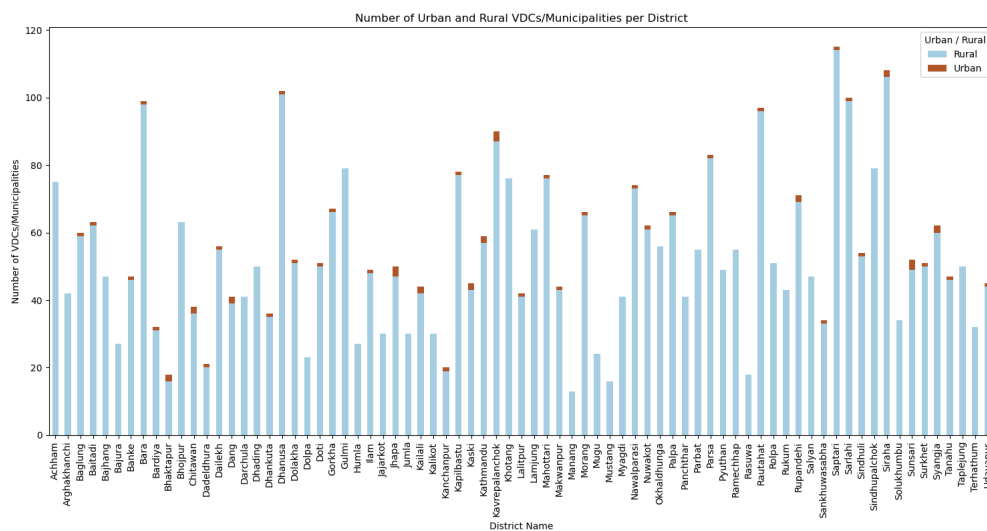
8.1. BatchId

BatchId provides the following comprehensive Insight into Administrative Hierarchy Mapping, Regional Disparity Analysis, Urban vs. Rural Trends, and Geo-stratified Analysis. The insights we extracted from the data are as follows:

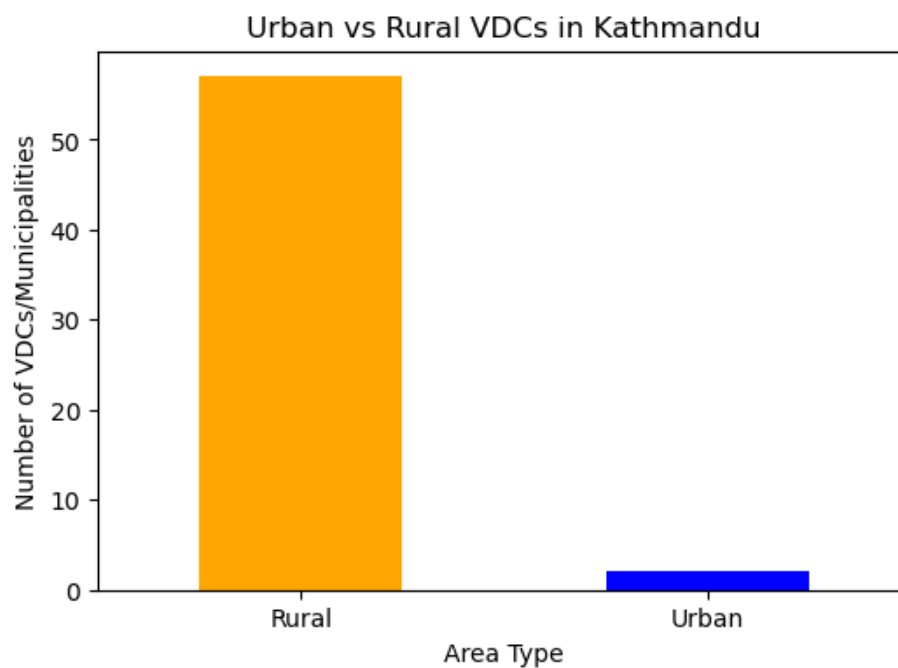
1. Number of VDCs/Municipalities in Each District



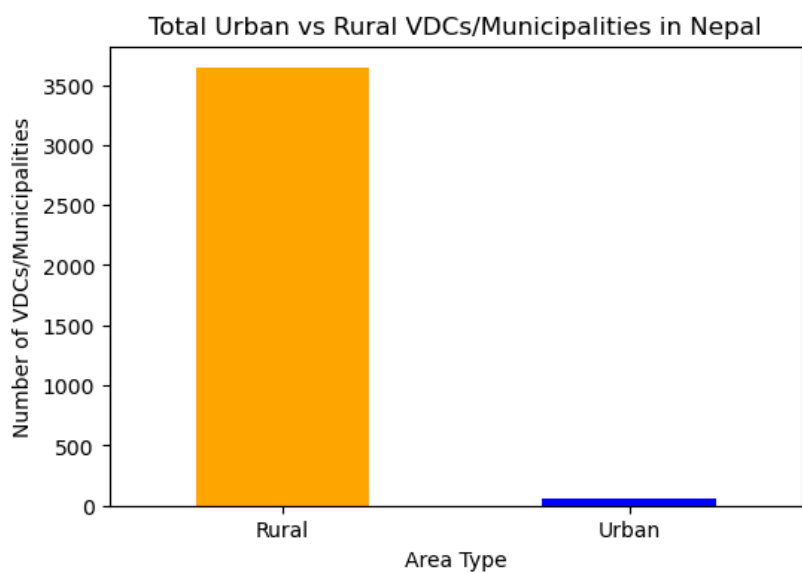
2. Number of Urban and Rural VDCs/Municipalities per District



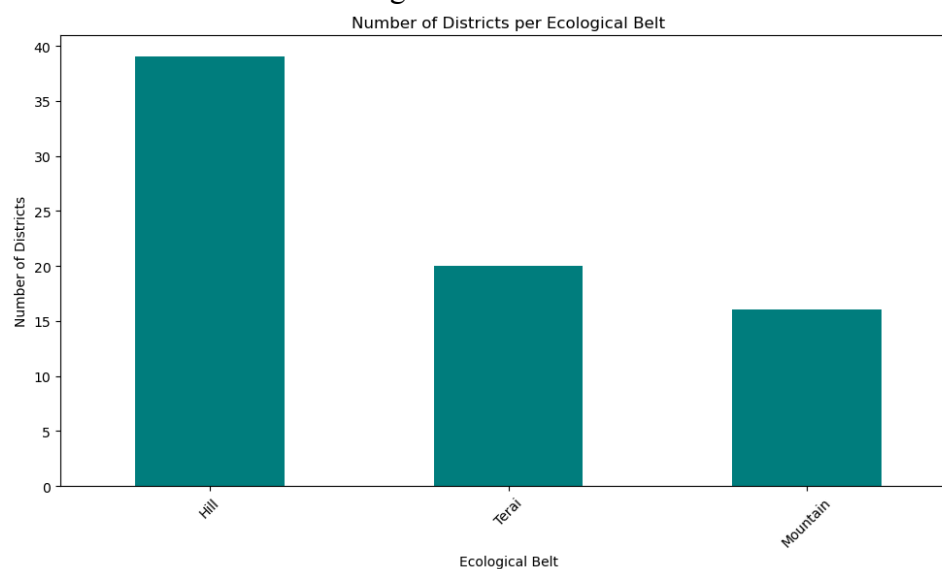
3. Urban vs Rural VDCs in Kathmandu



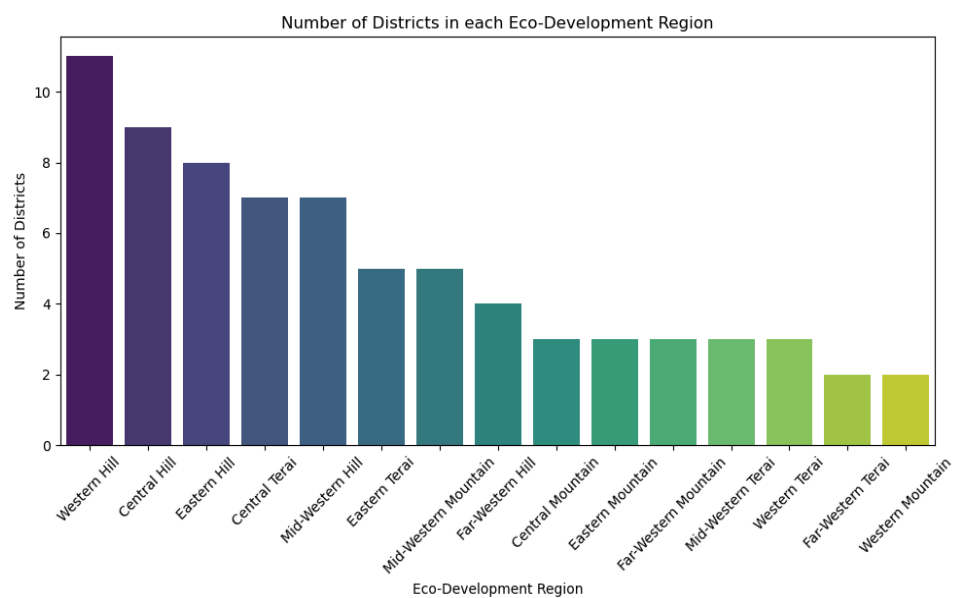
4. Total Urban Vs Rural VDCs/ Municipalities in Nepal



5. Number of Districts Per Ecological Belt

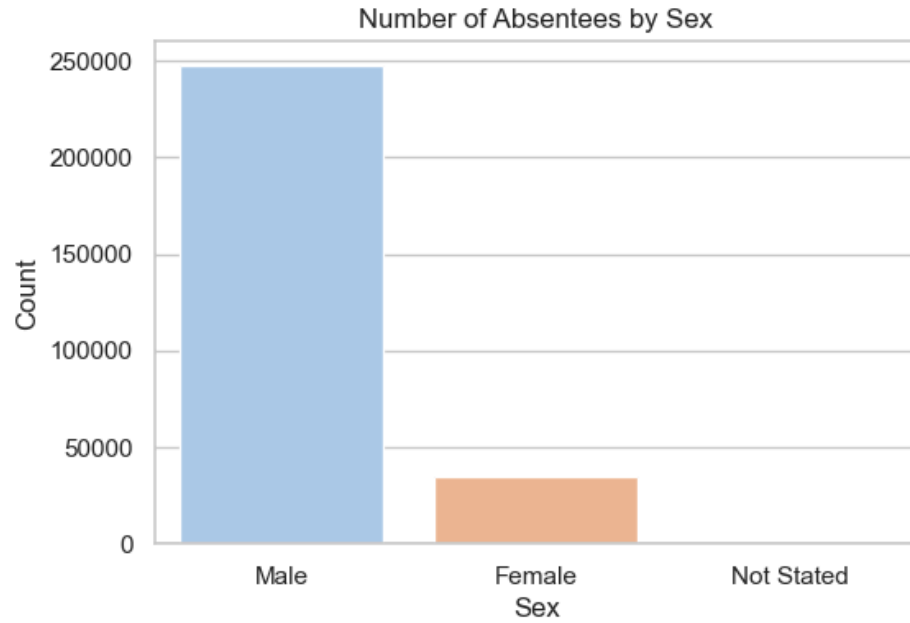


6. Number of Districts in each Eco-Development Region

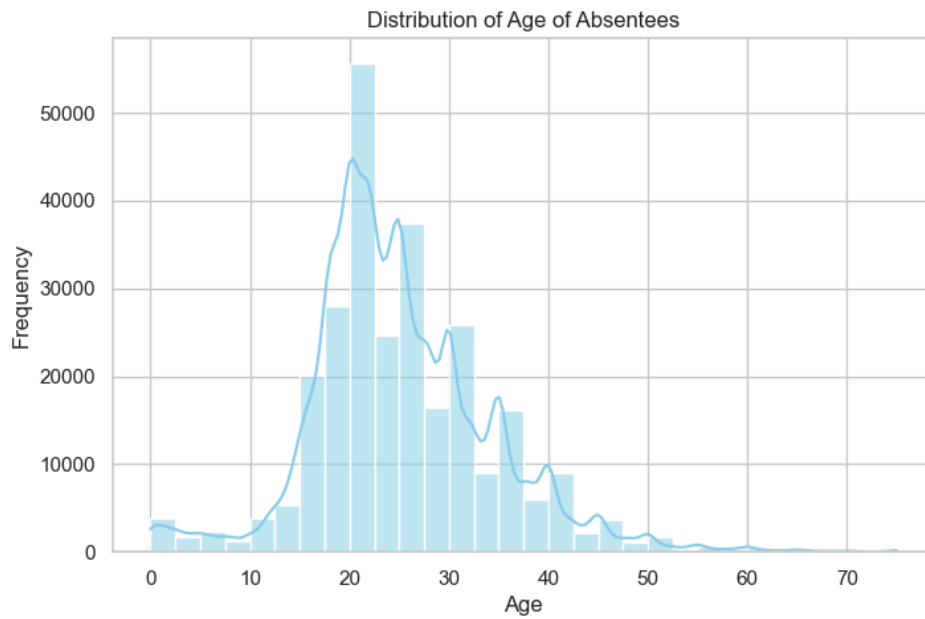


7. Urban Vs Rural VDCs/Municipalities per Eco-Development Region

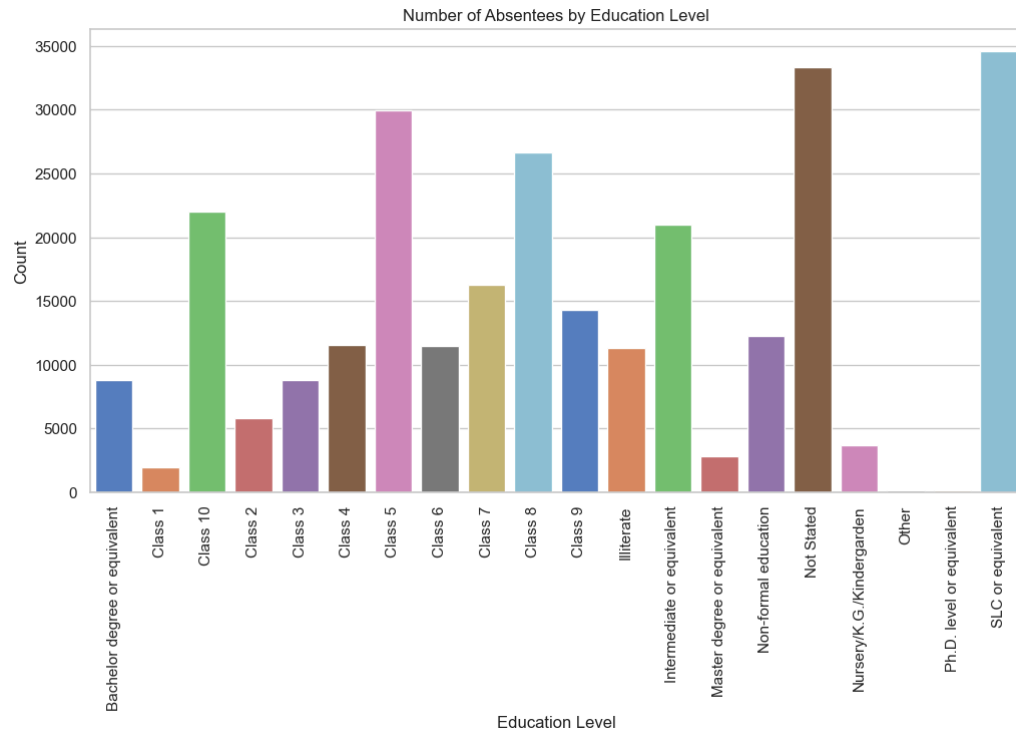
8.2. Absentee



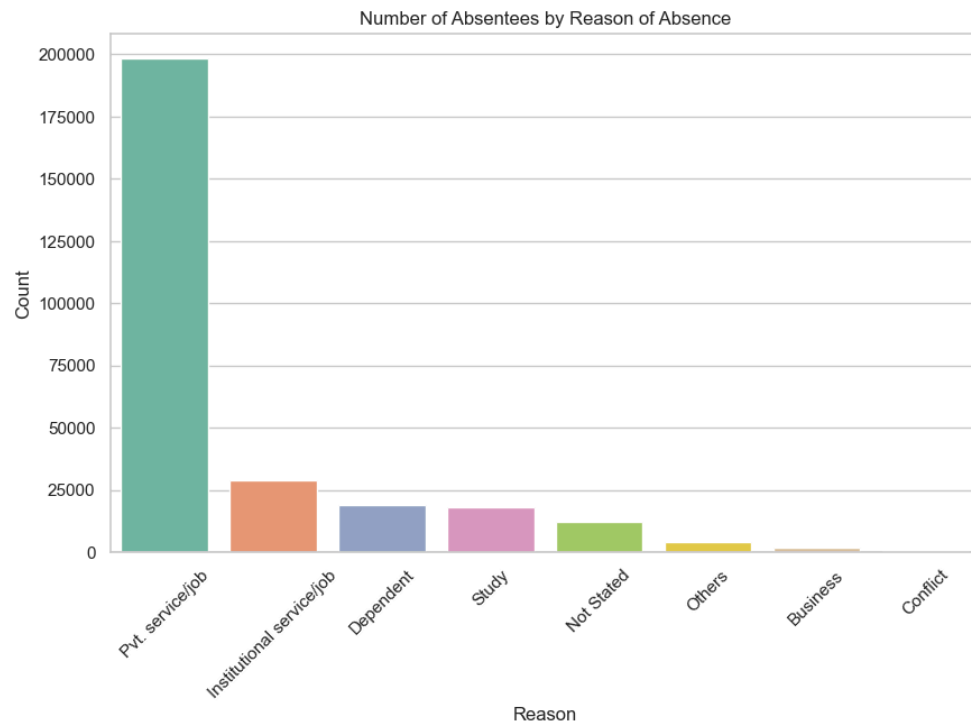
3. Distribution of Age of Absentees



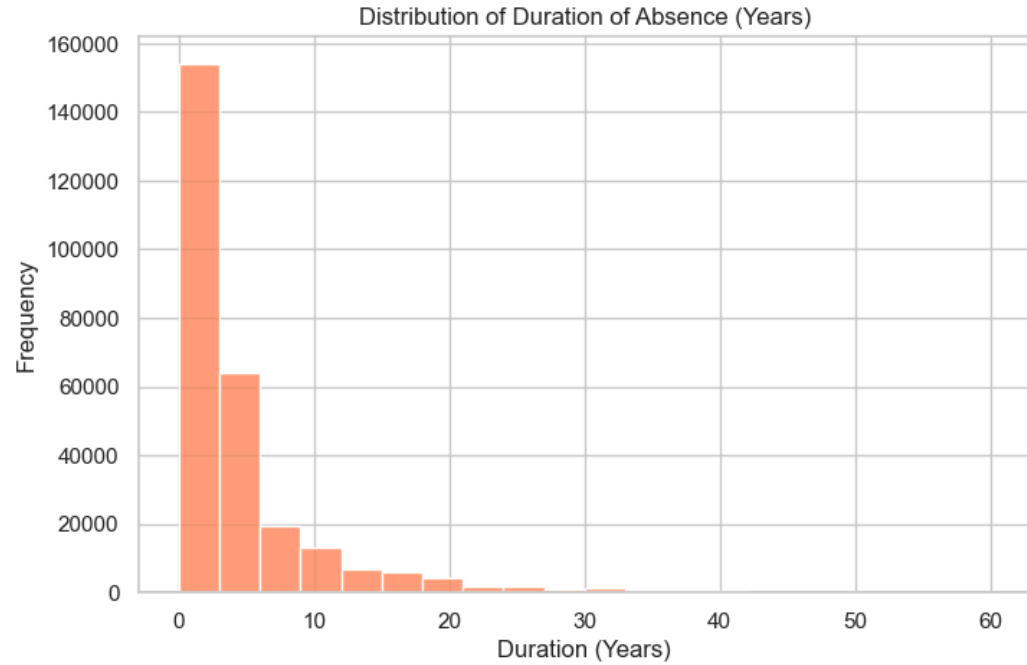
4. Number of Absentees by Education Level



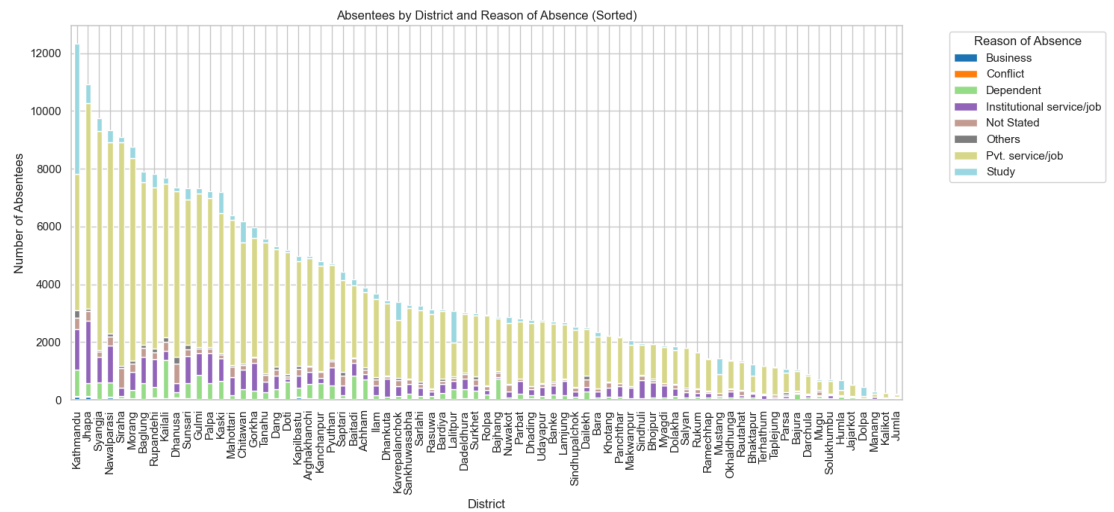
5. Number of Absentees by Reason of Absence



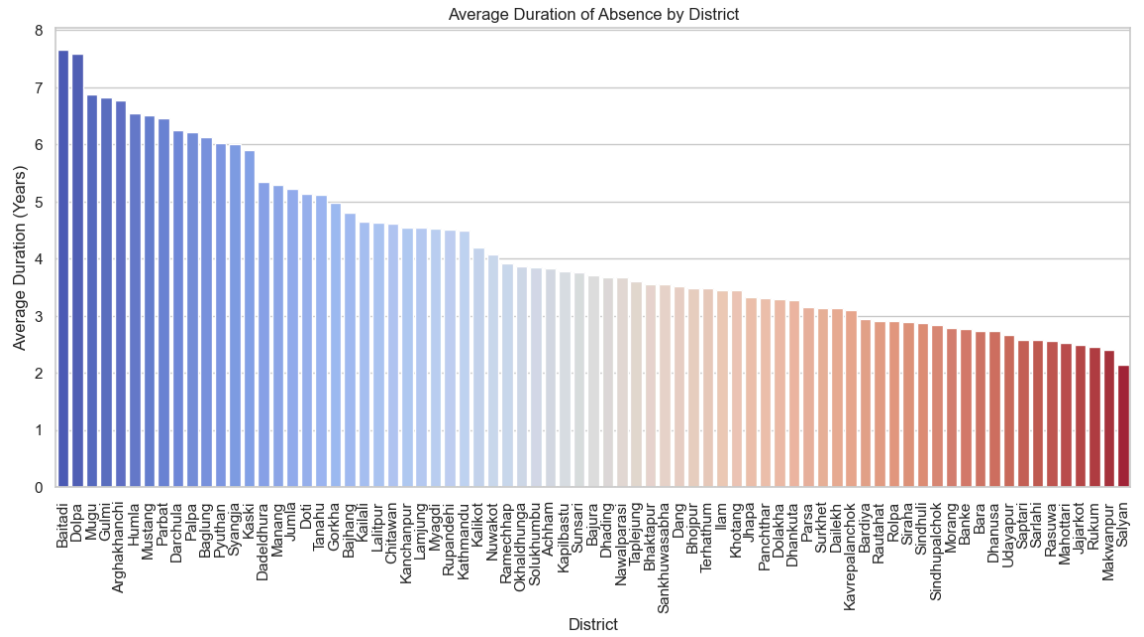
6. Distribution of Duration Absence (Years)



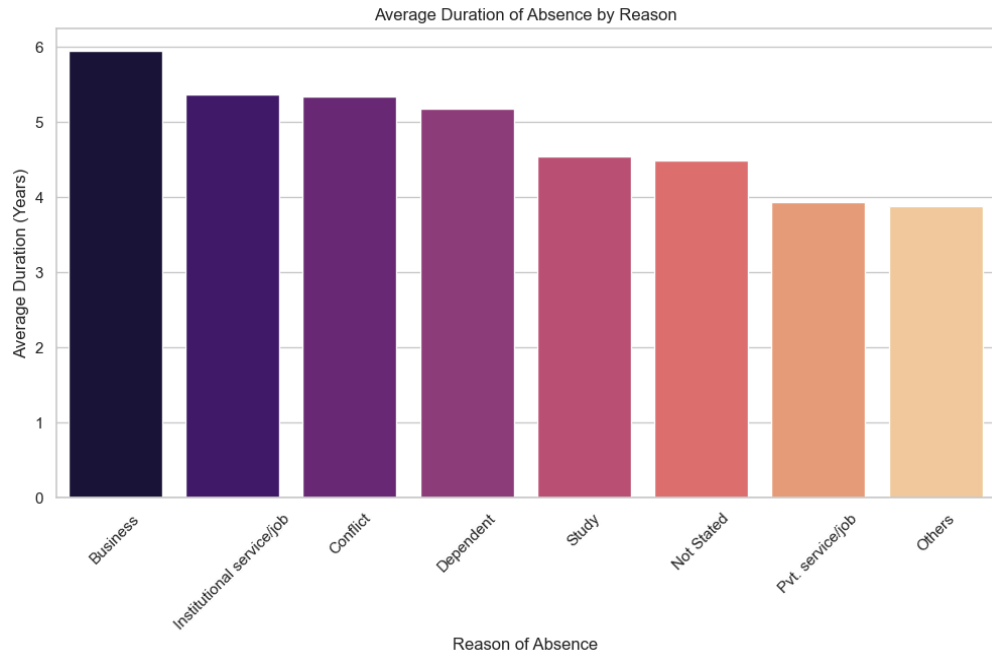
7. Absentees by District and Reason of Absence (Sorted)



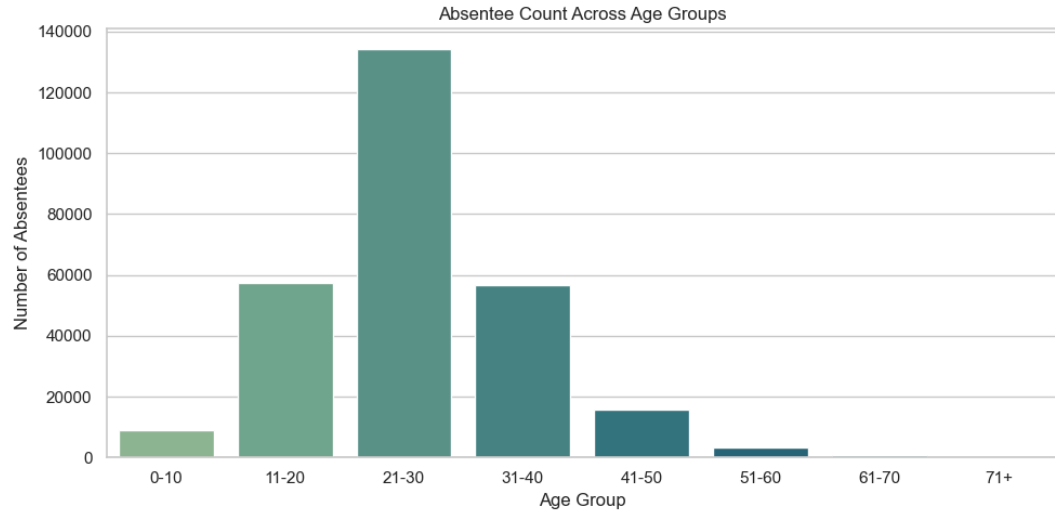
8. Average Duration of Absence by District



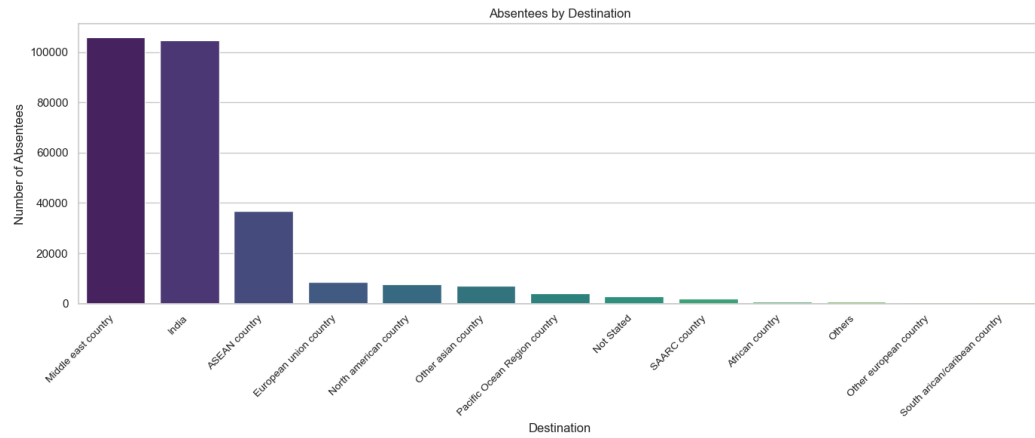
9. Average Duration of Absence by Reason



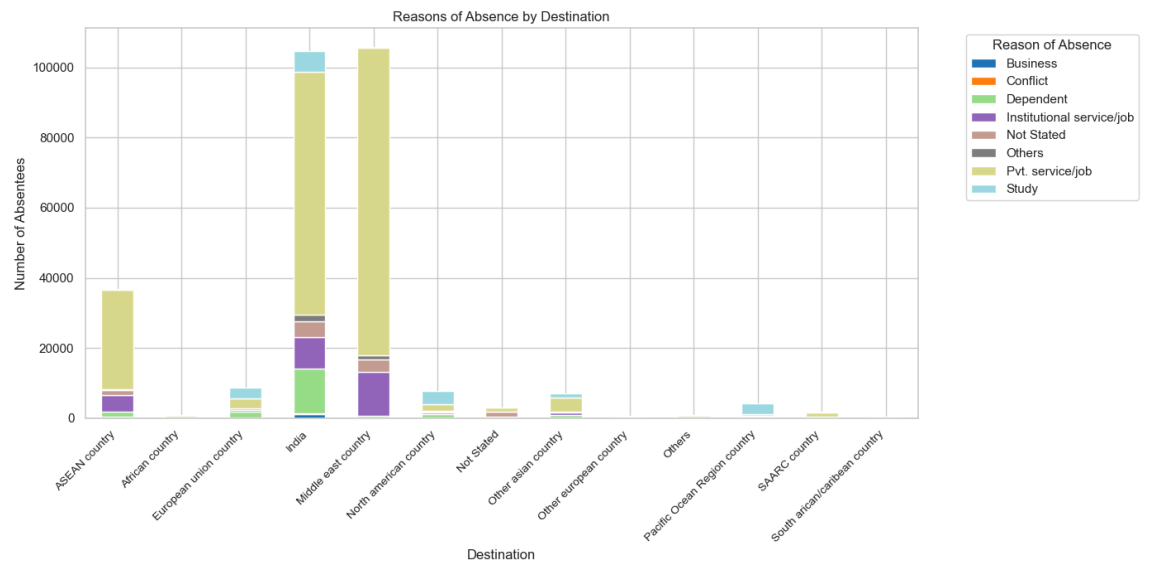
10. Absentee Count Across Age Group



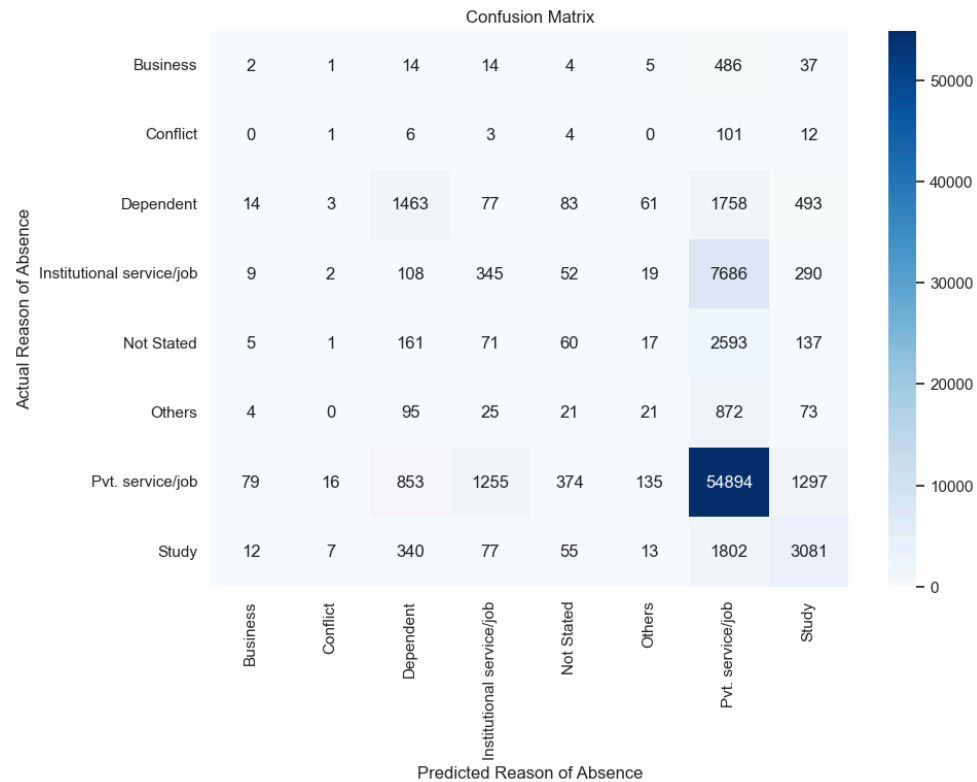
11. Absentees by Destination



12. Reason of absence by Destination



13. Confusion Matrix

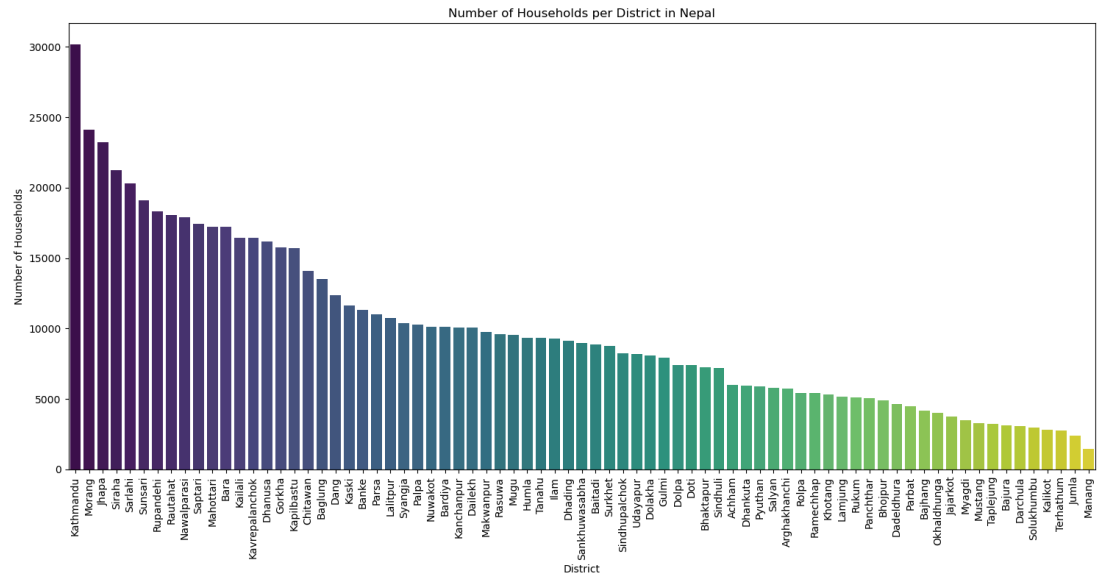


8.3. HouseHold

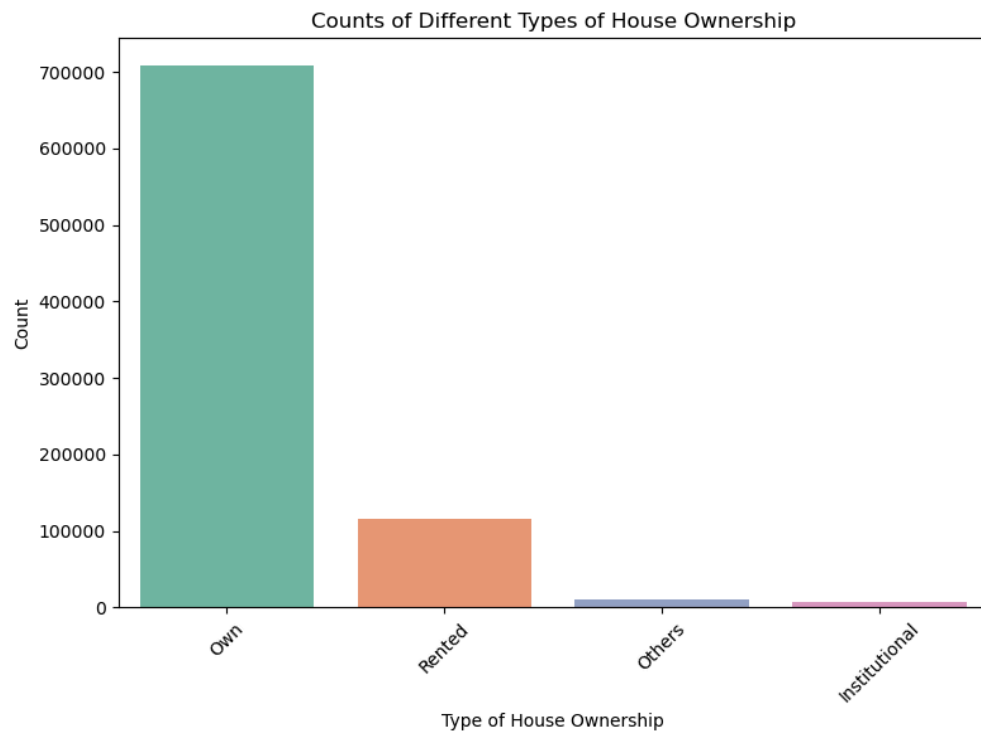
The household dataset provides detailed insights into living conditions and demographic patterns across Nepal at multiple administrative levels, including District, V.D.C./Municipality, Ward number, and Enumeration area. It captures information such as house and household numbers, types of house ownership, and structural characteristics like the foundation, outer wall, and roof types. Furthermore, it highlights socio-economic indicators such as female ownership of land—measured in local units (Bigha, Kattha, Dhur, Ropani, Aana, Paisa)—which is crucial for understanding gender-based property rights and economic empowerment. The dataset also records mortality within the last 12 months and the number of deceased, offering insight into public health and family structures. Additionally, the presence of absentee individuals abroad is documented, reflecting patterns of foreign employment and migration. Altogether, this comprehensive household information supports informed policy-making, targeted development

efforts, and a deeper understanding of community-level disparities across Nepal. The insights we extracted from the data are as follows:

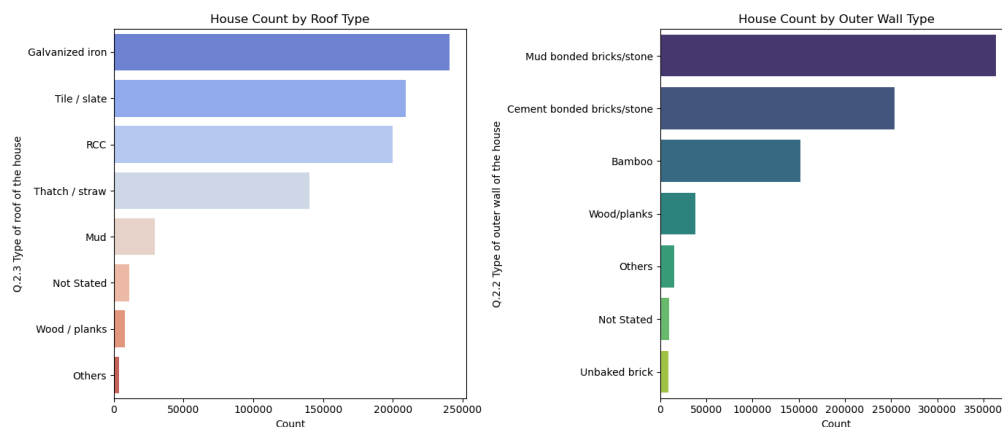
1. Number of Households per District in Nepal



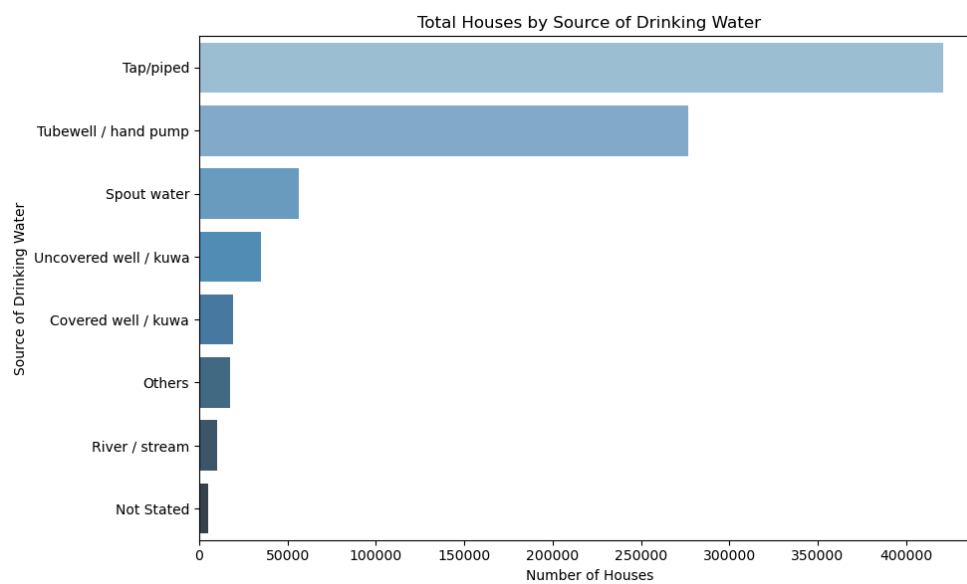
2. Count of Different Types of House Ownership



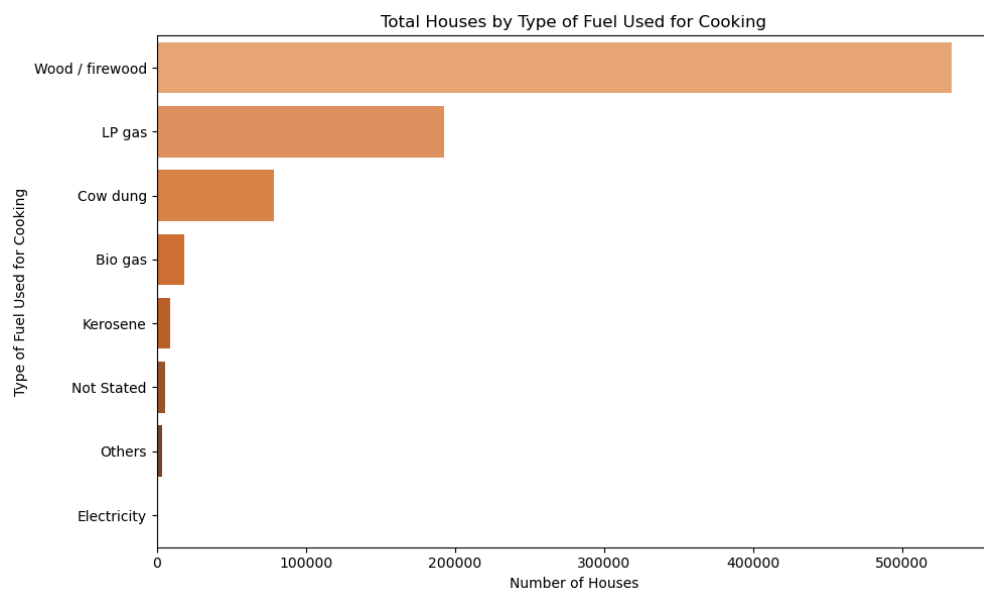
3. House count by roof / Outer Wall Type



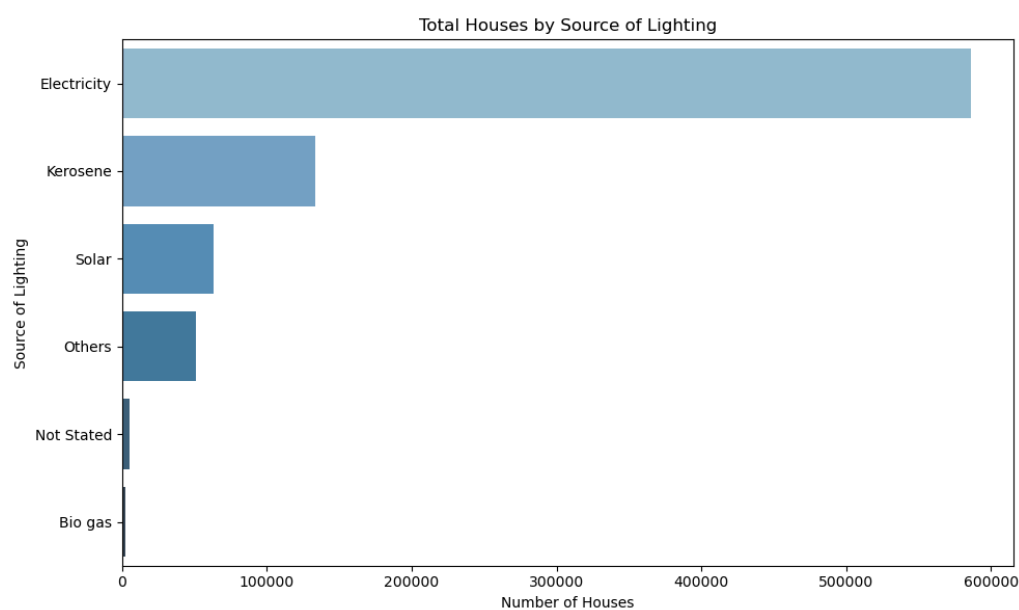
4. Total House by Source of Drinking Water



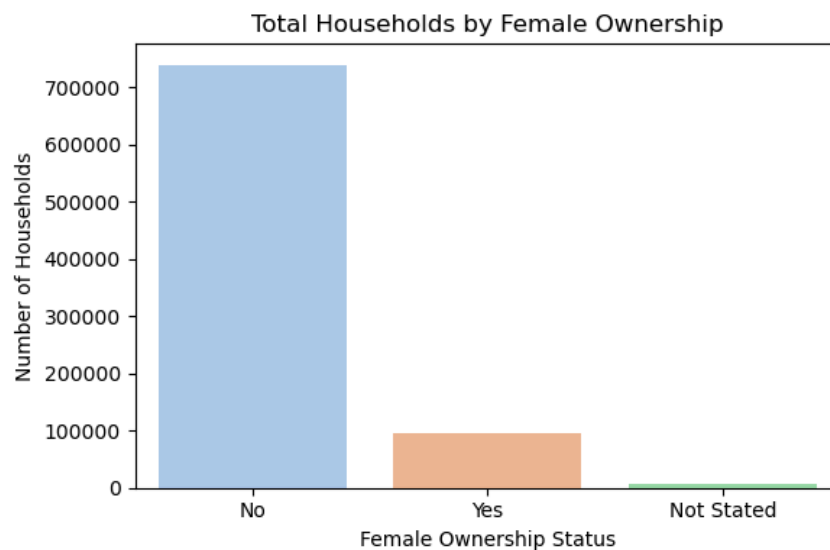
5. Total House by Type of Fuel Used for Cooking



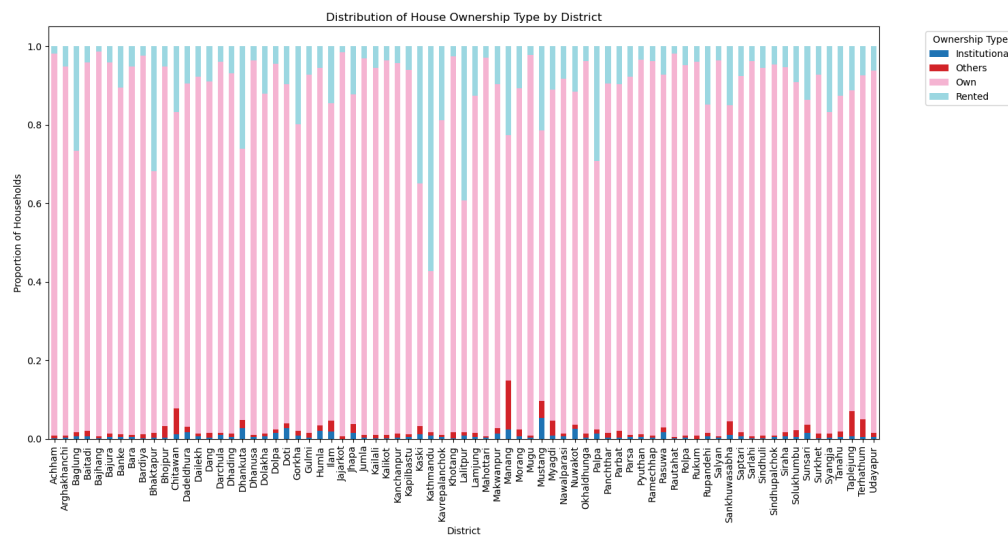
6. Total House by Source of Lighting



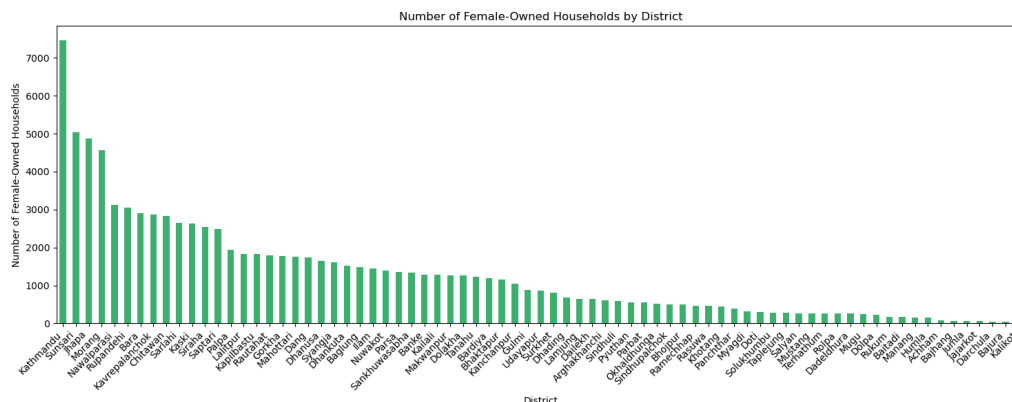
7. Total Households by Female Ownership



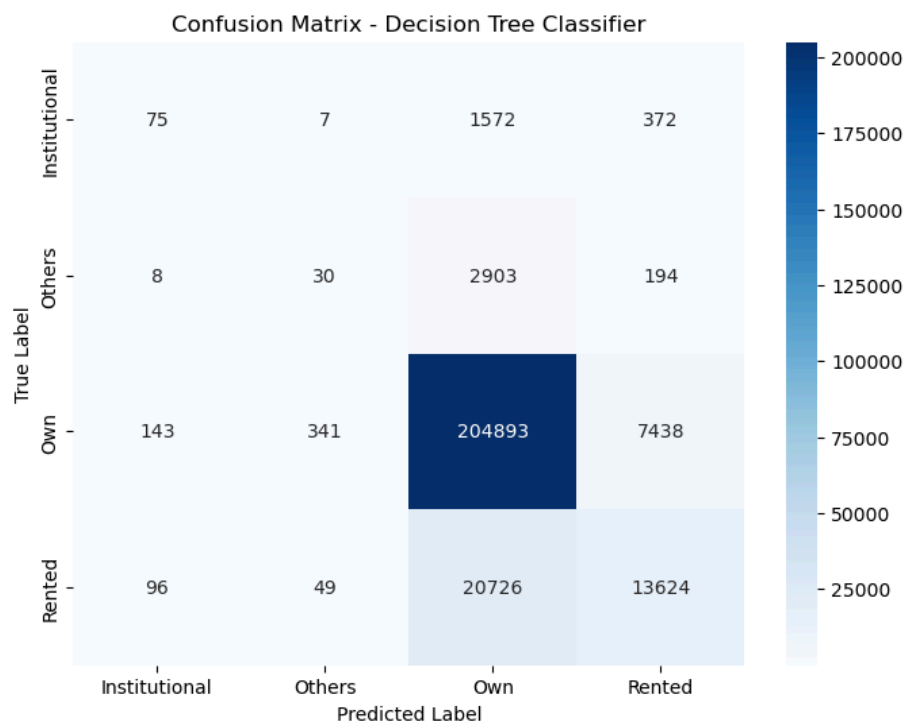
8. Distribution of House Ownership Type by District



9. Number of Female-Owned Households by District



10. Confusion Matrix - Decision Tree Classifier

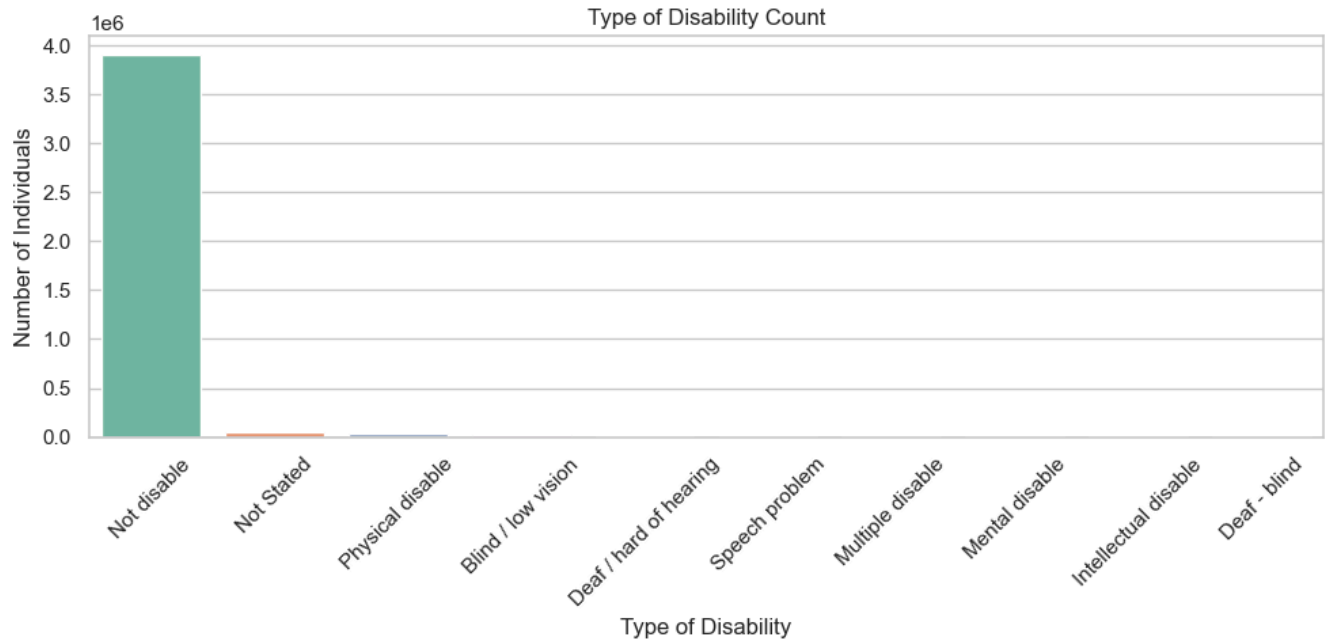


8.4. Individual

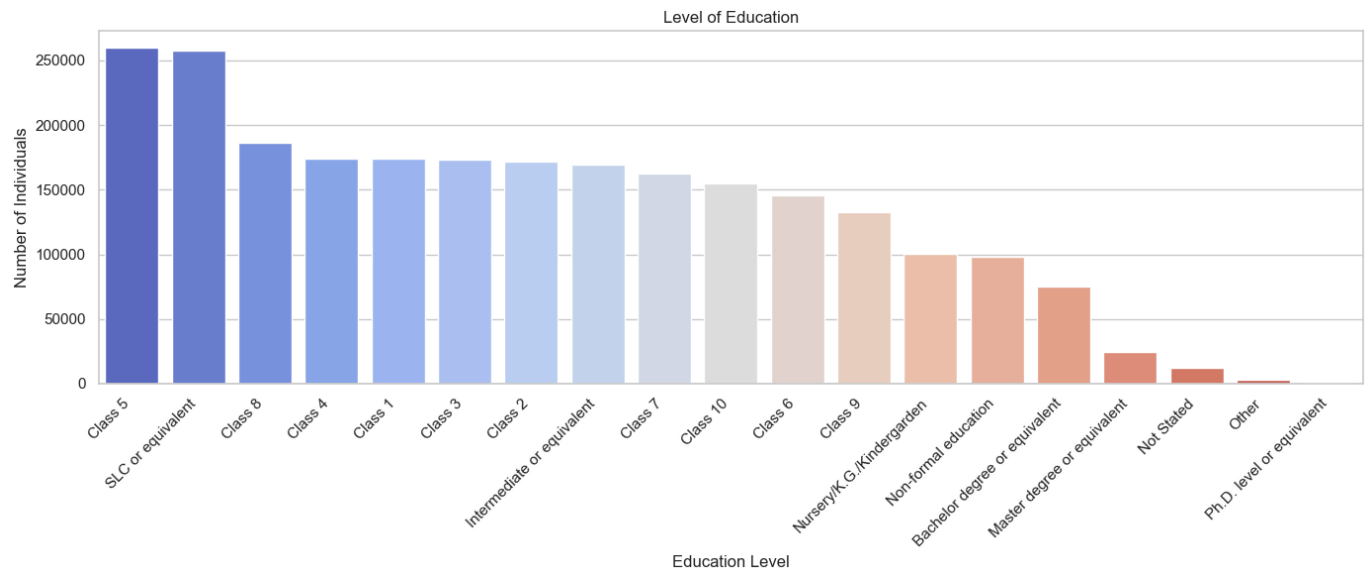
The individual dataset provides detailed demographic and socio-cultural information across Nepal, linking each person to a specific household through unique identifiers. It captures key attributes such as relationship to the household head, sex, age, age at first marriage, religion, mother tongue, second language, citizenship, and disability status. Educational indicators like literacy, school attendance, level, and sector of education offer insights into access and attainment.

This comprehensive data supports informed decision-making in areas such as education, health, inclusion, and social development.

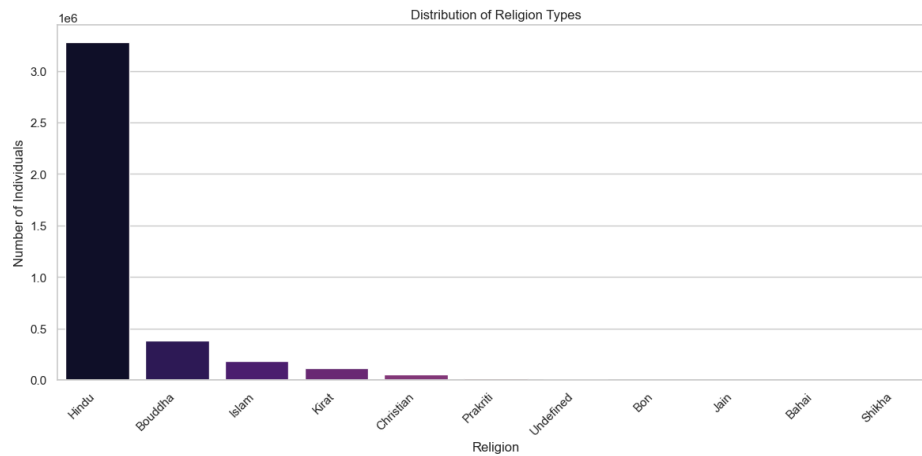
1. Type of Disability Count



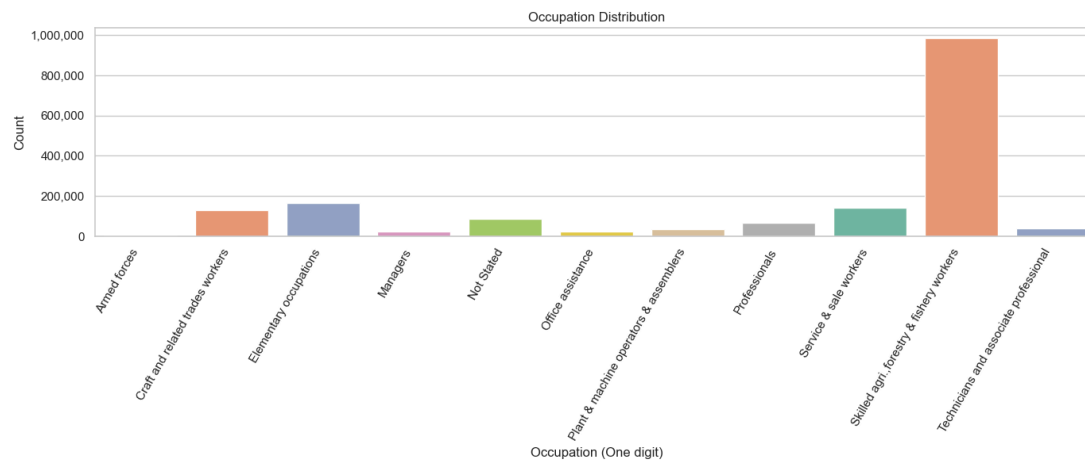
2. Level of Education



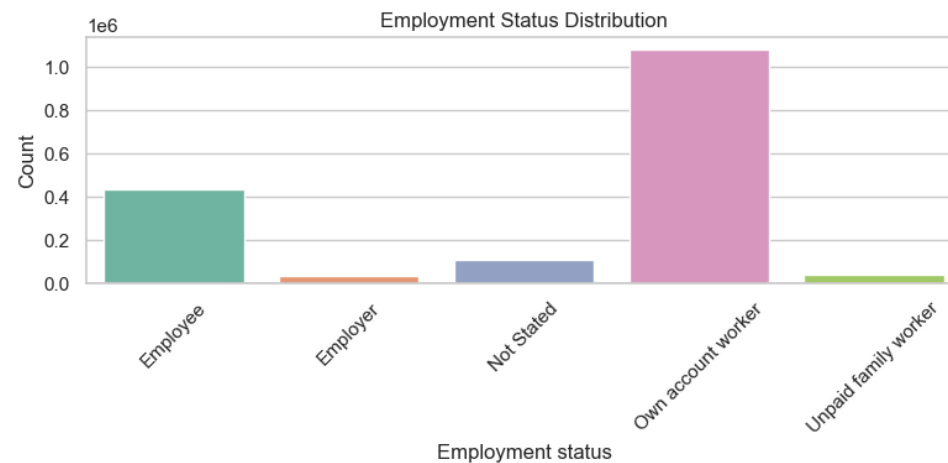
3. Distribution of Religion Types



4. Occupation Distribution



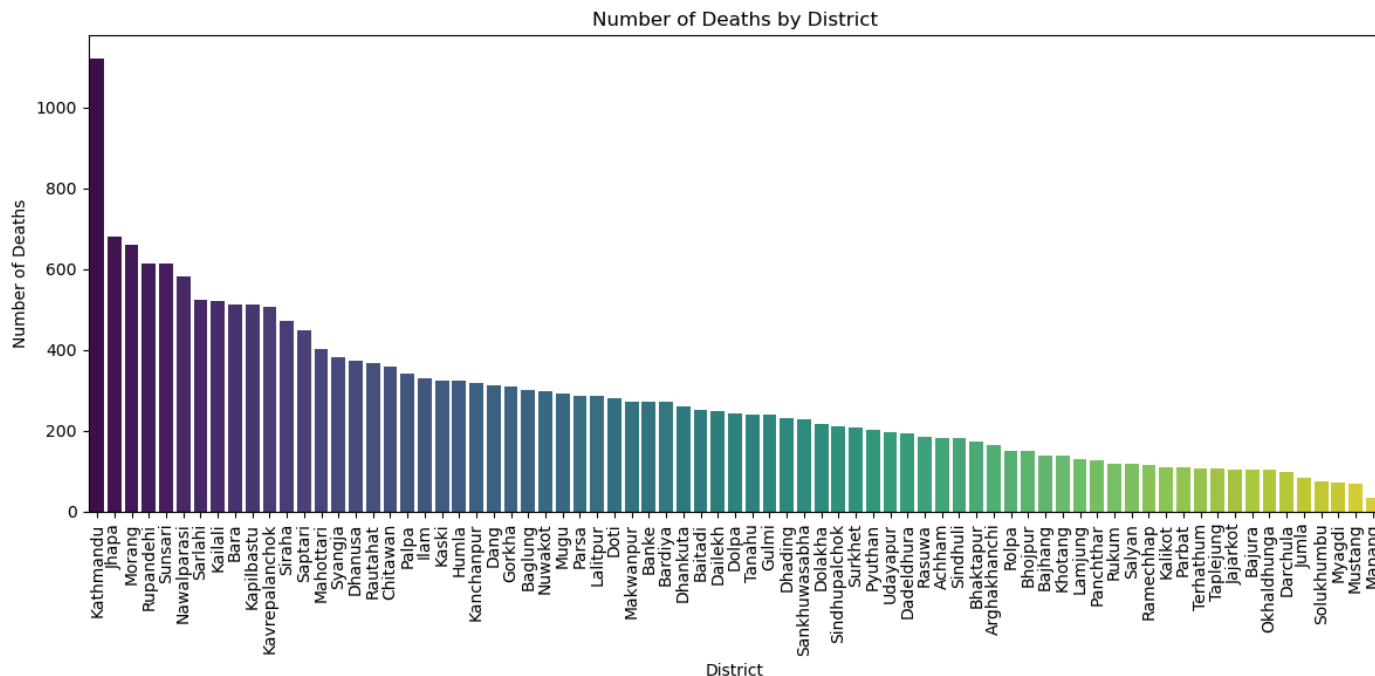
5. Employment Status Distribution



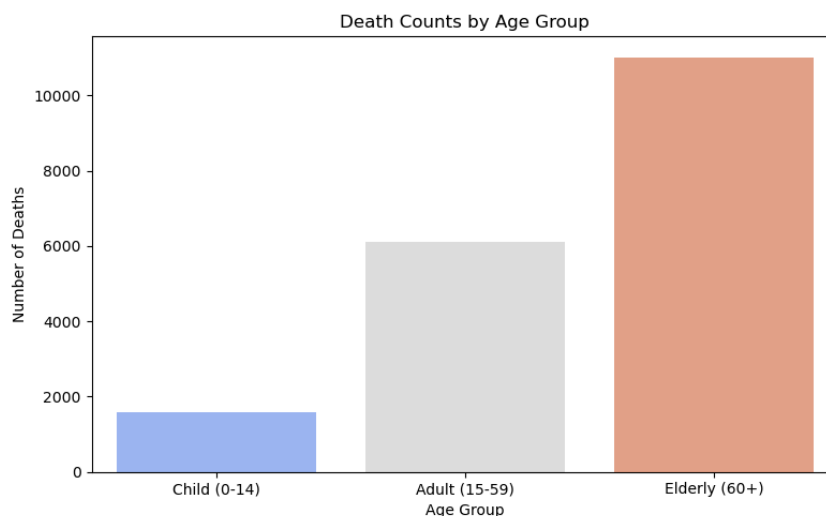
8.5. Death

The death dataset captures vital information on mortality across Nepal at granular administrative levels, including District, V.D.C./Municipality, Ward number, and Enumeration area. It records household and individual identifiers, along with the ID code, sex, and age of deceased individuals. This data is crucial for analyzing mortality patterns, assessing public health trends, identifying vulnerable age or gender groups, and guiding targeted health interventions and resource allocation.

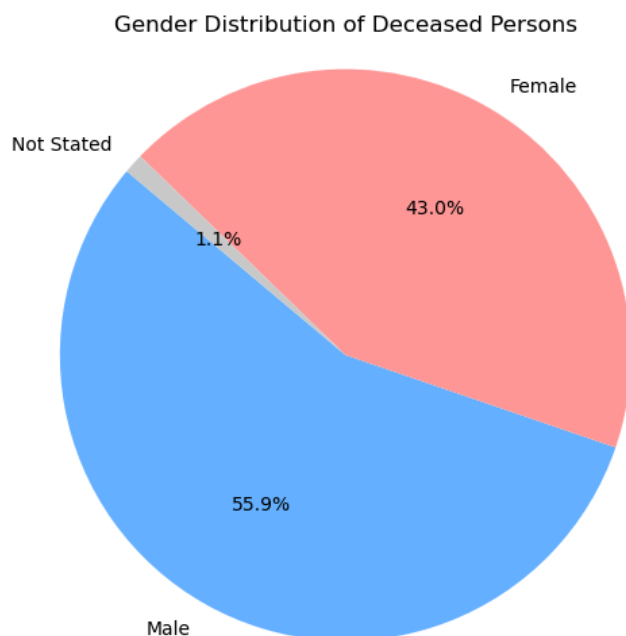
1. Number of Death by District



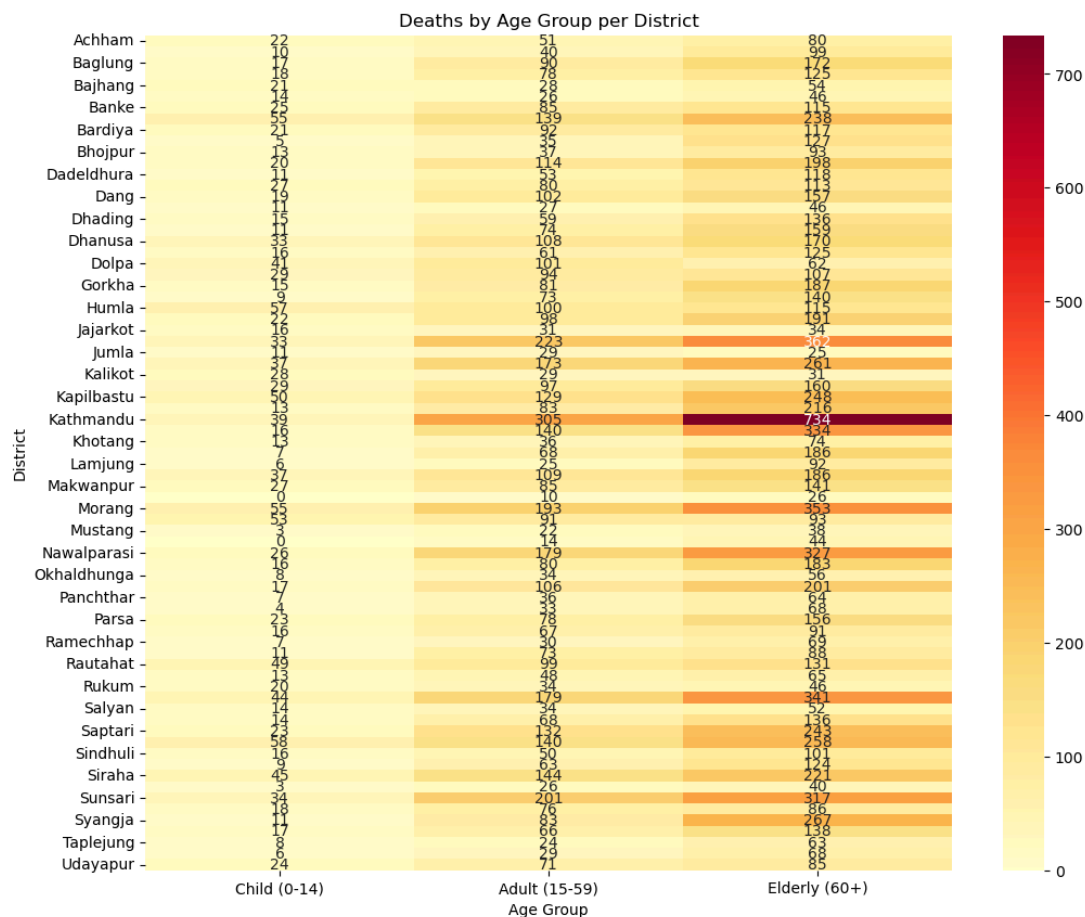
2. Death Count by Age Group



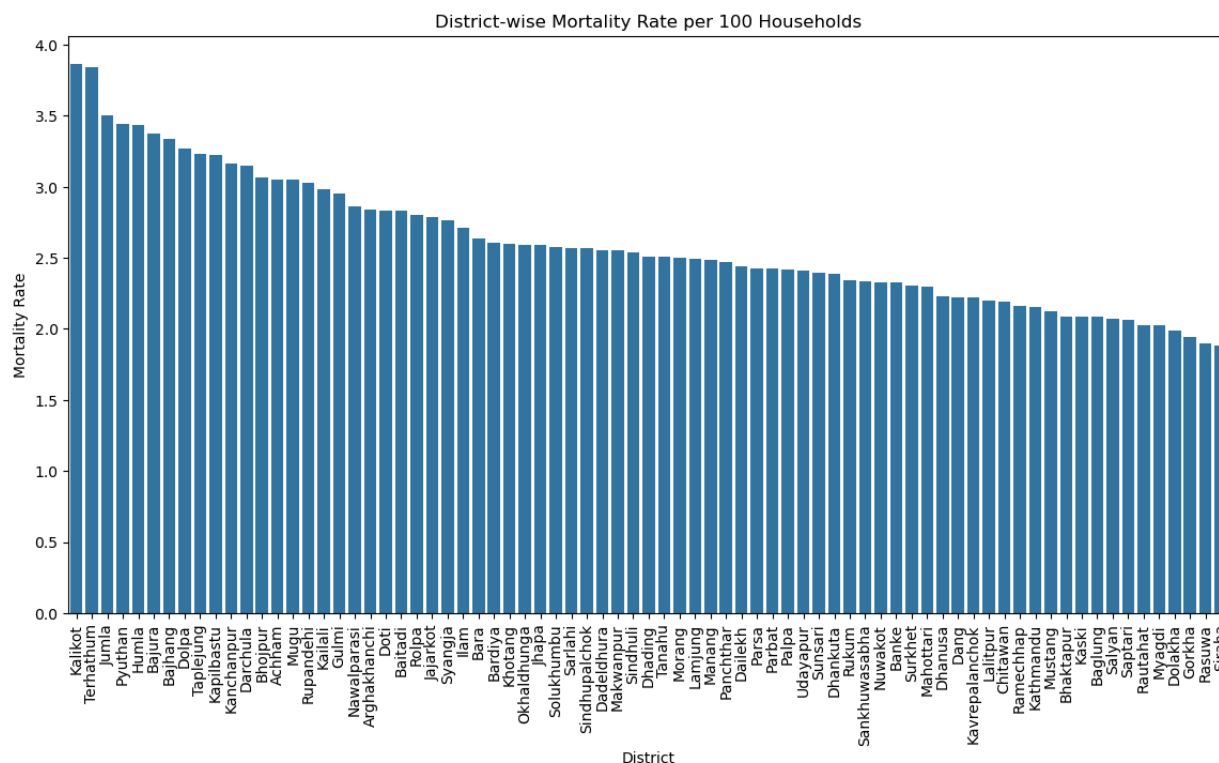
3. Gender Distribution of Deceased Persons



4. Death by age group per District



5. District-Wise Mortality Rate per 100 Households



8.6. Comparative Analysis

The 2011 Nepal Census marked a significant demographic milestone, recording a total population of approximately 26.5 million, which was an increase of over 3.3 million since the 2001 census, reflecting an annual growth rate of 1.35%. Compared to 2001, the average household size decreased from 5.44 to 4.88 persons, indicating a trend toward smaller family units. The working-age population (15-59 years) rose from 53% in 2001 to 57% in 2011, suggesting a demographic shift favorable for economic productivity and potential demographic dividend. Female-headed households increased notably from about 15% to nearly 26%, reflecting social changes. The census also detailed ethnic diversity, identifying 126 caste/ethnic groups, with Chhetri as the largest group at 16.6% of the population. The sex ratio declined from 99.8 males per 100 females in 2001 to 94.14 in 2011, indicating a relative increase in the female population. Population density increased from 157 to

181 persons per square kilometer, and Nepal entered the third stage of demographic transition, characterized by lower birth and death rates, contributing to slower population growth. Additionally, literacy and contraceptive use improved, reflecting rising education and awareness levels. Overall, the 2011 census reveals Nepal's demographic evolution with smaller households, a growing working-age population, increased female household leadership, and a slowing population growth rate, all pointing toward ongoing social and economic transformation.

9. Recommendations

The census emphasizes strengthening the census process for future rounds by building on the successes and lessons learned. Key suggestions include enhancing the quality and comprehensiveness of data collection by refining questionnaires to capture more detailed demographic, social, and economic variables. It is recommended to increase the use of technology in data collection and processing to improve accuracy and efficiency. Capacity building of census staff, particularly in training and management, is crucial to maintain high professional standards. The census also highlights the importance of inclusive workforce participation, encouraging greater involvement of women, marginalized ethnic groups, and backward classes to ensure representativeness and social equity in data gathering. Furthermore, sustained collaboration with development partners such as UNFPA, UNWOMEN, and others is advised to leverage technical and financial support. Finally, improving census publicity and community engagement is recommended to enhance public awareness and cooperation, which are vital for the success of such a large-scale statistical operation.

10. Conclusion

Data mining and data science provide powerful tools for understanding and improving agricultural commodity systems. For Nepal, these insights can guide interventions to boost productivity, reduce import dependency, and enhance export competitiveness. The roadmap outlined in this report demonstrates a replicable approach for other developing countries seeking to modernize their agricultural sector.

References

1. Central Bureau of Statistics. (n.d.). *National Population and Housing Census 2021 (Preliminary Results)*. Government of Nepal. Retrieved July 6, 20 25, from <https://censusnepal.cbs.gov.np/Home/Details?tpid=5&tfsid=1>
2. Wikipedia contributors. (2023, November 6). *Census in Nepal*. Wikipedia. https://en.wikipedia.org/wiki/Census_in_Nepal